

Topology General Exam

August 21, 2019

Instructions: This is a four hour exam. Your solutions should be legible and clearly organized, written in complete sentences in good mathematical style. All work should be your own-no outside sources are permitted-using methods and results from the first year topology course topics. Each problem is worth the same number of points.

Problem 1

Let $f : S^n \rightarrow S^n$ be a smooth map with the property that $df_x : T_x S^n \rightarrow T_{f(x)} S^n$ is injective for every $x \in S^n$.

- a) Prove that f is a diffeomorphism provided that $n \geq 2$.
- b) Find a counterexample to part (a) in the case that $n = 1$.

Problem 2

Let M be a connected smooth manifold of dimension m and let $N \subset M$ be a smooth submanifold of dimension $n = m - k$.

- a) Show that if $k \geq 2$ then $M \setminus N$ is connected.
- b) Prove that if $k \geq 3$ then the inclusion $M \setminus N \rightarrow M$ induces an isomorphism $\pi_1(M \setminus N, x_0) \rightarrow \pi_1(M, x_0)$ for $x_0 \in M \setminus N$ any basepoint.

Problem 3

Let M be a smooth compact manifold with nonempty boundary $N = \partial M$. Prove that there does not exist a retraction $M \rightarrow N$.

Problem 4

Let $M_n \cong \mathbb{R}^{n^2}$ denote the square $n \times n$ matrices with real entries.

- a) Prove that $\mathrm{SL}_n(\mathbb{R})$, the matrices with determinant 1, is a smooth submanifold of M_n .
- b) Prove that $\mathrm{SL}_n(\mathbb{R})$ is not compact.

Problem 5

Let K be a 2-complex consisting of one 0-cell, two 1-cells labelled $\{a, b\}$, and a single 2-cell attached along the loop $aba^{-1}b$.

- a) Find $H_*(K; \mathbb{Z})$.
- b) Find $H_*(K; \mathbb{Z}/2\mathbb{Z})$.
- c) Show that K is homeomorphic to a smooth manifold.

Problem 6

Let $f : S^n \rightarrow S^n$ be a continuous map.

- a) Prove that if f has no fixed points then f is homotopic to the antipodal map.
- b) Let G be a group acting freely on S^n , where here n is even. Prove that G has order at most two. [Hint: construct a homomorphism to $\mathbb{Z}/2\mathbb{Z}$ using degree and argue that this map is injective.]

Problem 7

Let X be the 1-point union of two copies of the projective plane, $X = \mathbb{R}P^2 \vee \mathbb{R}P^2$.

- a) Find a presentation for $\pi_1(X)$ and describe $H_1(X; \mathbb{Z})$ as a direct sum of cyclic groups.
- b) Prove that every continuous map $f : X \rightarrow S^1$ is nullhomotopic.

Problem 8

Let C be a circle that is smoothly embedded in $\mathbb{R}^3$. For example C could be the trefoil knot:



A single closed loop is drawn with three crossings in the familiar three-lobed trefoil projection. At each crossing, a gap marks the strand that passes underneath. The diagram is an example of a smoothly embedded circle; the question explicitly asks for a method that applies to any such circle.

but your methods should not be specific to this case. Find the homology groups $H_* (\mathbb{R}^3 \setminus C; \mathbb{Z})$.